

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 217 — Past Questions & Materials
by Renance

Course: MTH 217
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

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Paper 1 of 2 — MTH 217 Real Analysis First Semester 2020 20210

Level 200 • First Semester • PastQuestion

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UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2020/2021 ACADEMIC SESSION
MTH 217: REAL ANALYSIS I. Time Allowed: 2hrs
Instruction: Answer Four (4) Questions Only

QUESTION ONE

a) Find all $x \in \mathbb{R}$ that satisfy the following inequalities (i) $|4x - 3| \leq 11$ (ii) $|x^2 - 4| < 5$ (6mks)
 b) Show that for all $p, q \in \mathbb{R}$ (i) $pq \leq \frac{1}{2}(p^2 + q^2)$ (ii) $\left(\frac{p+q}{2}\right)^2 \leq (p^2 + q^2)$ (6 ½ mks)
 c) Prove that if p and q are any two real number then (i) $p < q$ iff $p - q < 0$ (ii) if $p < q$ and $q < r$ then $p < r$. (5 mks)

QUESTION TWO

a) Define the following terms clearly (i) l.u.b (ii) g.l.b (iii) modulus of $\mathbb{R}$ (5 ½ mks)
 b) Hence or otherwise, prove the uniqueness of l.u.b and g.l.b (6mks)
 c) Find the l.u.b and g.l.b if they exists of the following sets: (i) $\left\{\frac{3n+2}{2n+1}, n \in \mathbb{N}\right\}$ (ii) $\left\{1 - \frac{1}{n}, n \in \mathbb{N}\right\}$
 (iii) $\left\{\frac{1}{5n}, n \in \mathbb{Z}, n \neq 0\right\}$ (6 mks)

QUESTION THREE

a) Justify whether or not the following sets are bounded. (i) $\mathbb{N}$ (ii) $\mathbb{Z}$ (iii) $S = (0,1)$ (iv) $I = (a, \infty)$ (v) $S = \left\{\frac{n}{2n+1}, n \geq 1\right\}$ (vi) $S = \{(-1)^n, n \geq 1\}$ (6mks)
 b) Find the distance between the real number $\alpha = -3$ and the sets (i) $T = \{0,1,2\}$ (ii) $S = \{0,1\}$ (iii) $R = \{1,2\}$ (6mks)
 c) Prove by induction that $\sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$. (5 ½ mks)

QUESTION FOUR

(a) Prove that $\lim_{n \rightarrow \infty} \left(\frac{2n-1}{3n+2}\right) = \frac{2}{3}$ (6 mks)
 (b) Verify using quotient method that the sequences defined by (i) $a_n = \frac{5n}{n+4}$ and (ii) $b_n = \frac{2n+7}{3n+8}$ are monotonic. (6 mks)
 (c) Let $b_1 = 1$ and $b_{n+1} = \sqrt{2 + b_n}$. Show that $\{b_n\}_{n=1}^{\infty}$ is convergent and find the limit. (5 ½ mks)

QUESTION FIVE

(a) Define clearly the terms (i) subsequence of a real number (ii) limit inferior (iii) limit superior (6 mks)
 (b) Define Cauchy sequence of a real number. Prove that a convergent sequence of real number is a Cauchy sequence (6 mks)
 (c) Hence or otherwise, show that the sequence given by $a_n = 5 + \frac{1}{n^2}$ is Cauchy (5 ½ mks)

QUESTION SIX

(a) Use $\epsilon - \delta$ definition to prove that $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by (i) $f(x) = 3x^2 - 2x + 1$ is continuous at $x = 5$ (6 mks)
 (b) When is a function said to be differentiable on an interval? (ii) Let $f: \mathbb{R} \setminus \{5\} \rightarrow \mathbb{R}$ be given by $f(x) = \frac{1}{x-5}$. Show that f is differentiable on $\mathbb{R} \setminus \{5\}$. (5 ½ mks)
 (c) Find the infimum and supremum of the following functions (i) $f: (2,3) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{x-2}$ (ii) $f: [-2,2] \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x}{2x+x^2}$ (iii) $f: (-2,2) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{x-2}$. (6mks)

Best wishes!!! $\|x_{n+1} - p\|^2 \leq \|x_n - p\|^2 - 2\langle x_n - p, x_n - p \rangle = \|x_n - p\|^2 - 2\|x_n - p\|^2 = -\|x_n - p\|^2$
 $= \|x_n - p\|^2 - 2\|x_n - p\|^2 = -\|x_n - p\|^2$
 $= \|x_n - p\|^2 - 2\|x_n - p\|^2 = -\|x_n - p\|^2$

Paper 2 of 2 — MTH 217: REAL ANALYSIS

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS/COMPUTER SCIENCE

FIRST SEMESTER EXAMINATION FOR 2017/2018 ACADEMIC SESSION

MTH 217: REAL ANALYSIS I TIME ALLOWED: 2HRS:30MINS

INSTRUCTION: ANSWER FOUR (4) QUESTIONS ONLY.

QUESTION ONE

- Find all $x \in \mathbb{R}$ that satisfy the following inequalities (i) $|4 - 3x| \leq 11$ (ii) $|x - 4| < 5$
- Show that for all $x, y \in \mathbb{R}$ (i) $x \leq (x + y)$ (ii) $x \leq (x + y)$
- Prove that if x and y are any two real number then (i) $x < y$ iff $-x > -y$ (ii) if $x < y$ and $y < z$ then

QUESTION TWO

- Define the following terms clearly (i) greatest integer function (ii) bounded set (iii) least upper bound (iv) greatest lower bound
- Justify whether or not the following sets is bounded. (i) $\mathbb{R}$ (ii) $[0, 1]$ (iii) $(0, 1)$ (iv) $(-\infty, \infty)$ (v) $x, y \geq 1$ (vi) $\{(-1)^n, n \geq 1\}$
- Prove that (i) $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$ (ii) $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

QUESTION THREE

- State and prove Bernoulli's inequality
- Verify that the sequence defined by $x_n = 1 + \frac{1}{n}$ (i) is monotone (ii) $\lim_{n \rightarrow \infty} x_n = e$ where e stands as the Euler number.
- Let $x_n = 1$ and $y_n = 2 + \frac{1}{n}$. Show that $\{x_n\}$ is convergent and find the limit.

QUESTION FOUR

- State the necessary condition for a series to be convergent.
- Verify the convergence or otherwise of the following series using limit comparison test (i) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ (ii) $\sum_{n=1}^{\infty} \frac{1}{n}$ (iii) $\sum_{n=1}^{\infty} \frac{1}{n^3}$
- Apply Ratio Test on the following series (i) $\sum_{n=1}^{\infty} \frac{1}{n!}$ (ii) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ to show its convergence or otherwise

QUESTION FIVE

- State the following theorems without proof (i) Rolle's theorem (ii) Mean Value theorem
- Find $c \in (0, 1)$, when the Mean Value theorem is applied to $f(x) = (x - 1)(x - 2)$
- If $f(x) = x^2, f(0) = 0$. Show that $f'(0)$ exists and find it.

QUESTION SIX

- Use $\epsilon - \delta$ definition to prove that $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 3 - 2x + 1$ is continuous at $x = 2$

- (b) When is a function said to be differentiable on an interval? (ii) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^2 \sin\left(\frac{1}{x}\right)$. Show that f is differentiable on $\mathbb{R}$.
- (c) Find the infimum and supremum of the following functions (i) $f : (2,3] \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{x}$ (ii) $f : [-2,2] \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ (iii) $f : (-2,2) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{x}$.